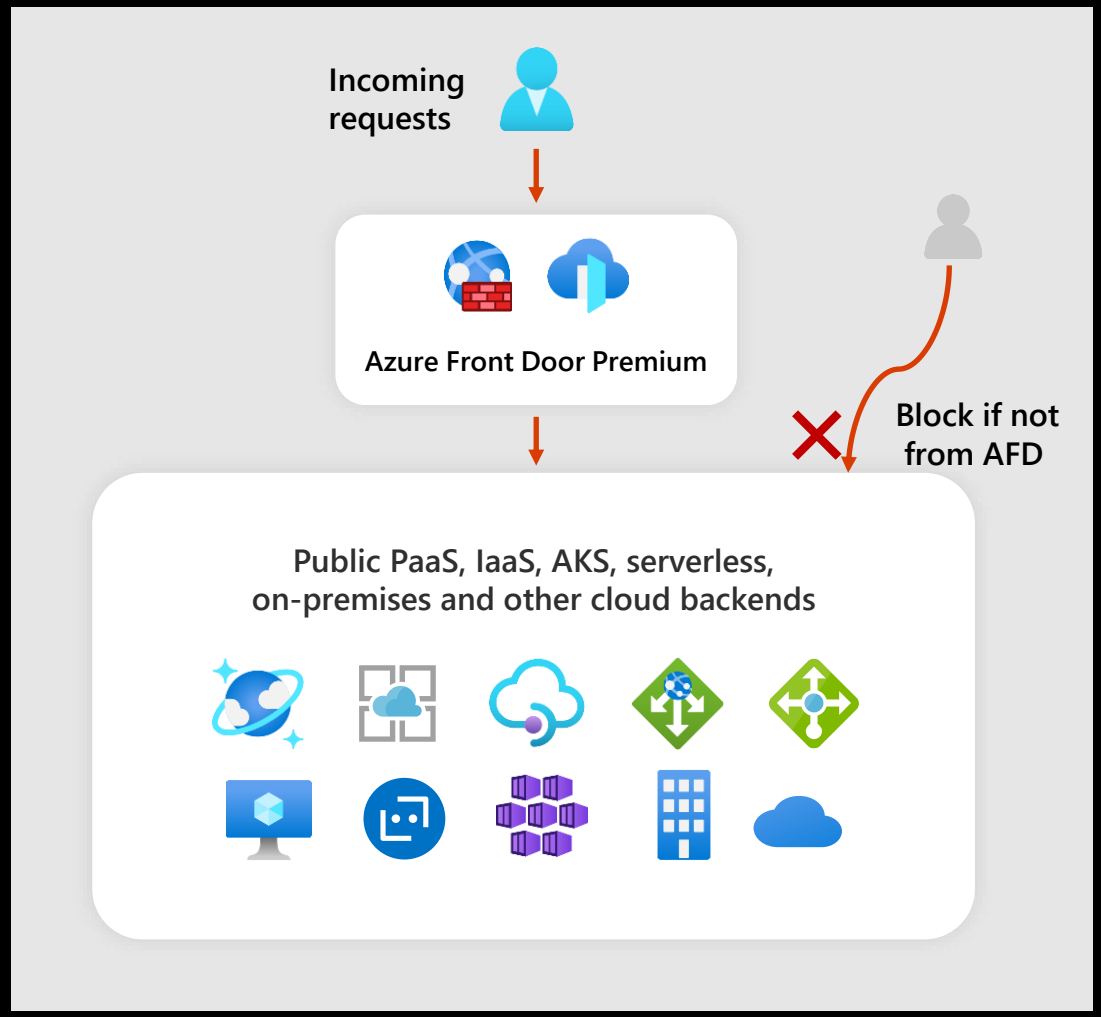
Azure Front Door

Global Layer 7 Load Balancer

with WAF and CDN



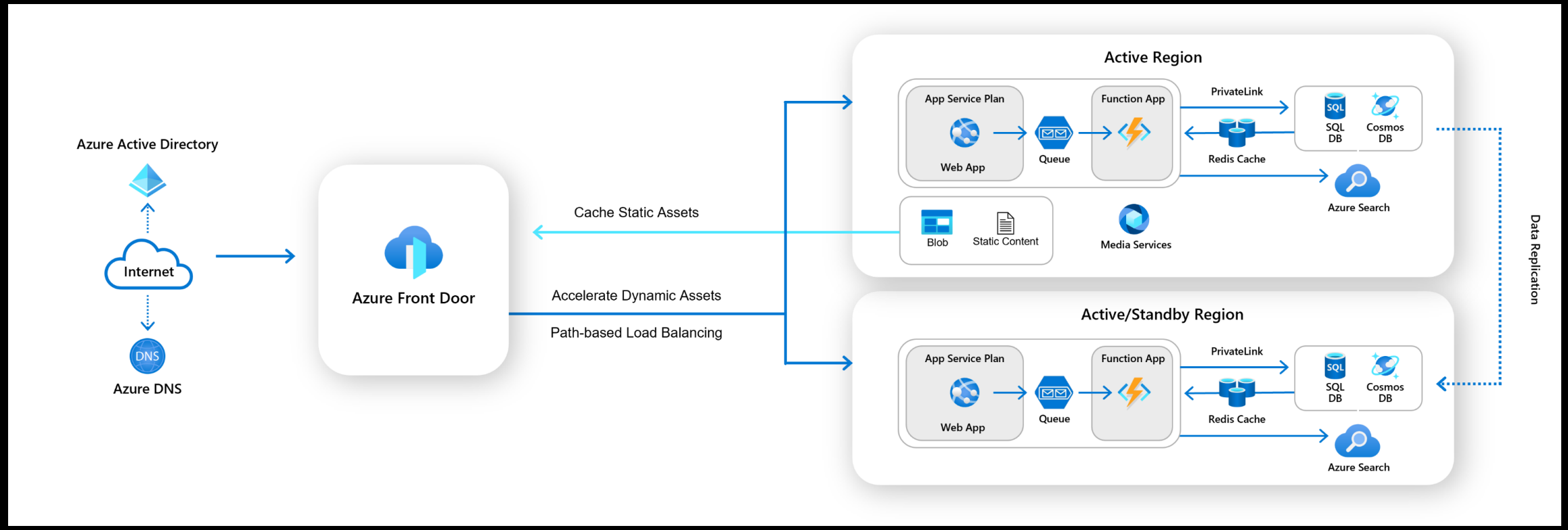
Front Door vs Azure load balancing services

- Global Load Balancer
- Layer 7 (HTTP/S)
- TLS offloading
- CDN
- WAF

	Application Gateway	Front Door	Load Balancer	Traffic Manager
Supported protocols	HTTP, HTTPS, HTTP2	HTTP, HTTPS, HTTP2	TCP, UDP	Any
Private load balancing	✓		✓	
Global load balancing		✓	✓	✓
Supported environments	Azure, non-Azure cloud, on prem	Azure, non-Azure cloud, on prem	Azure	Azure, non-Azure cloud, on prem
Host and path based load balancing	✓	✓		
TLS offloading	✓	✓		
Site acceleration		✓		
Security	WAF, NSG	WAF	NSG	

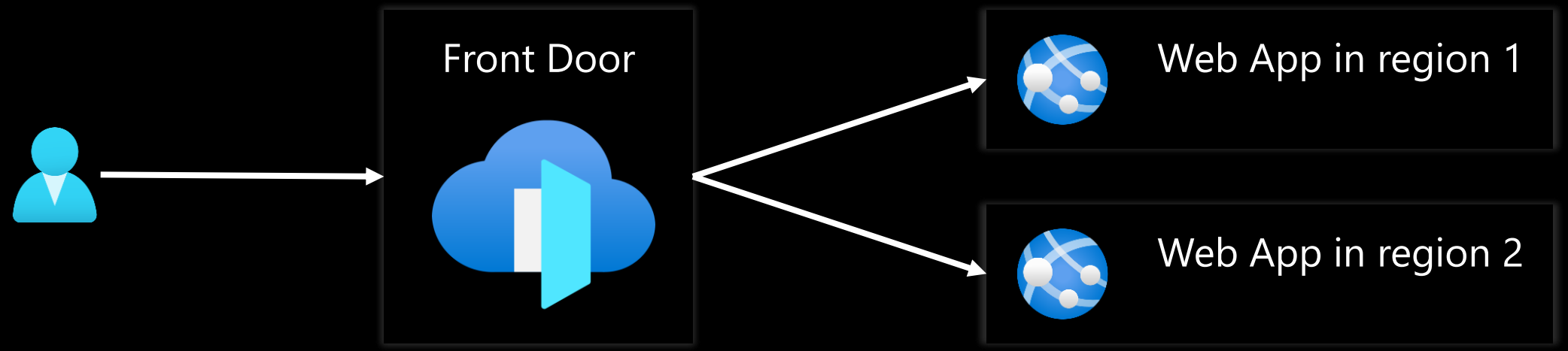
Front Door routes traffic to multiple regions

Front Door can load balance the traffic to applications deployed in multiple Azure regions or on-prem. This is useful for architectures using the following patterns: Active/Active, Active/Passive, Blue/Green and global applications.



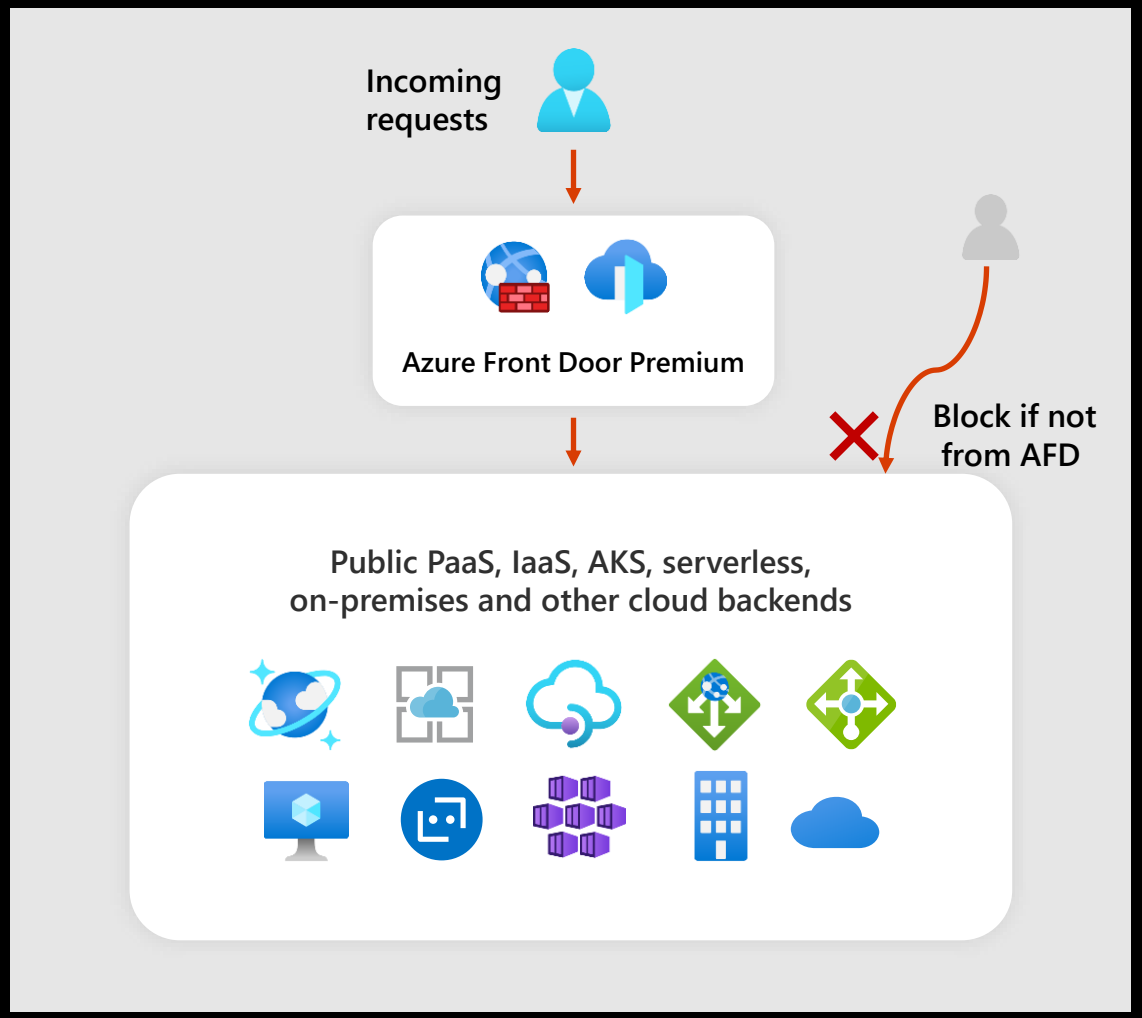
Exposing App Services using Front Door

Front Door can route traffic to multiple public endpoints like App Services, even within different regions.

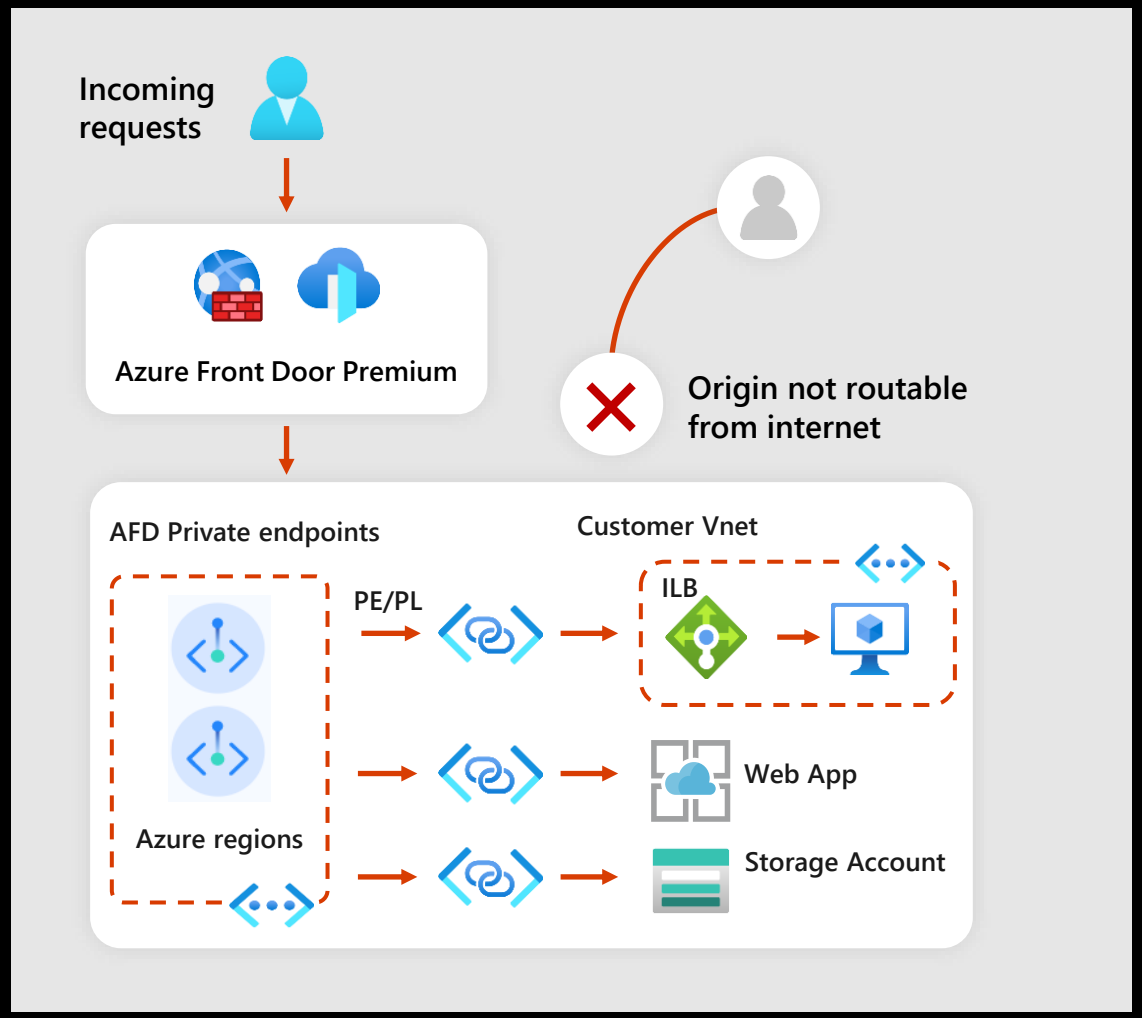


Front Door for public and private services

Public origins via IP and header restriction



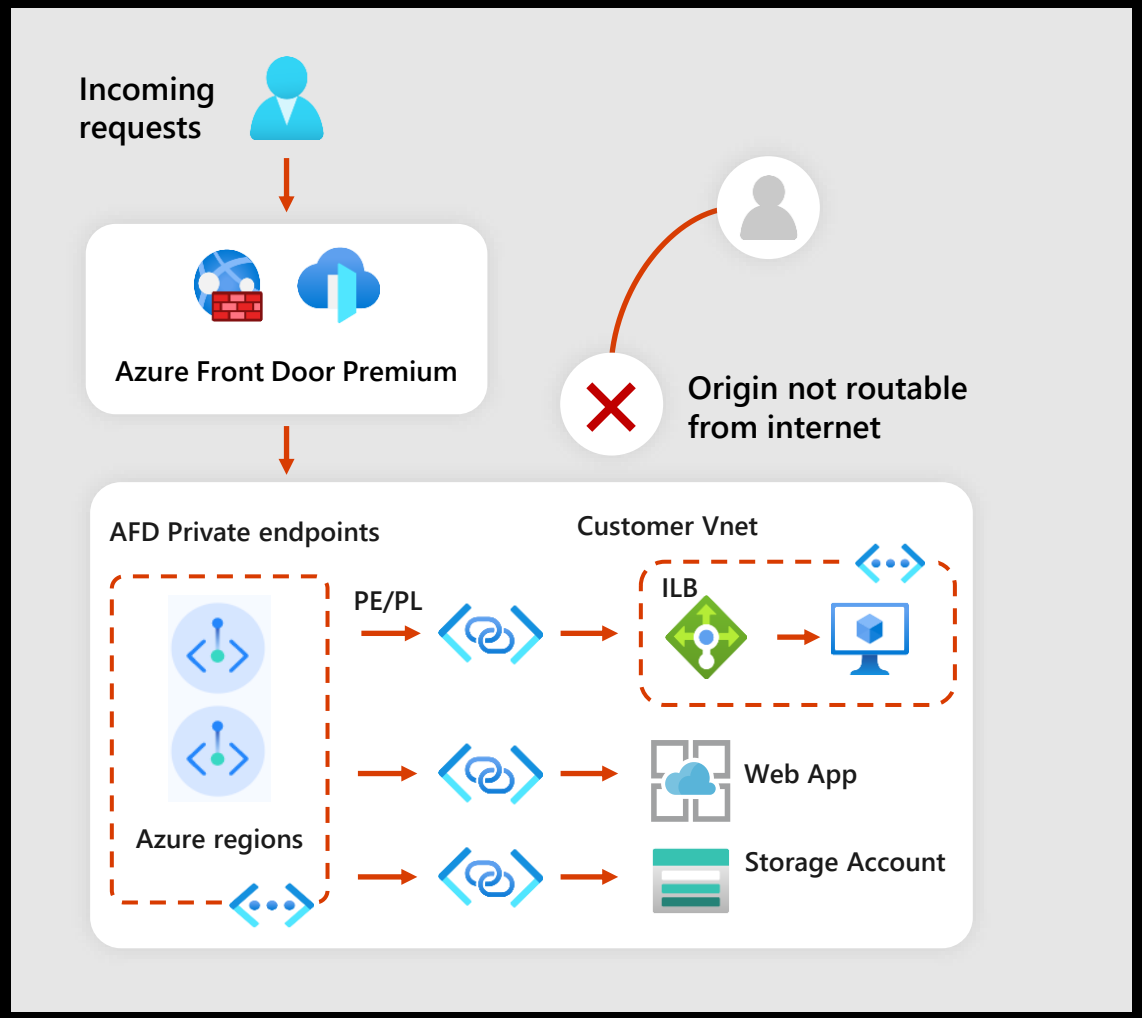
Private origins in Azure via private link service



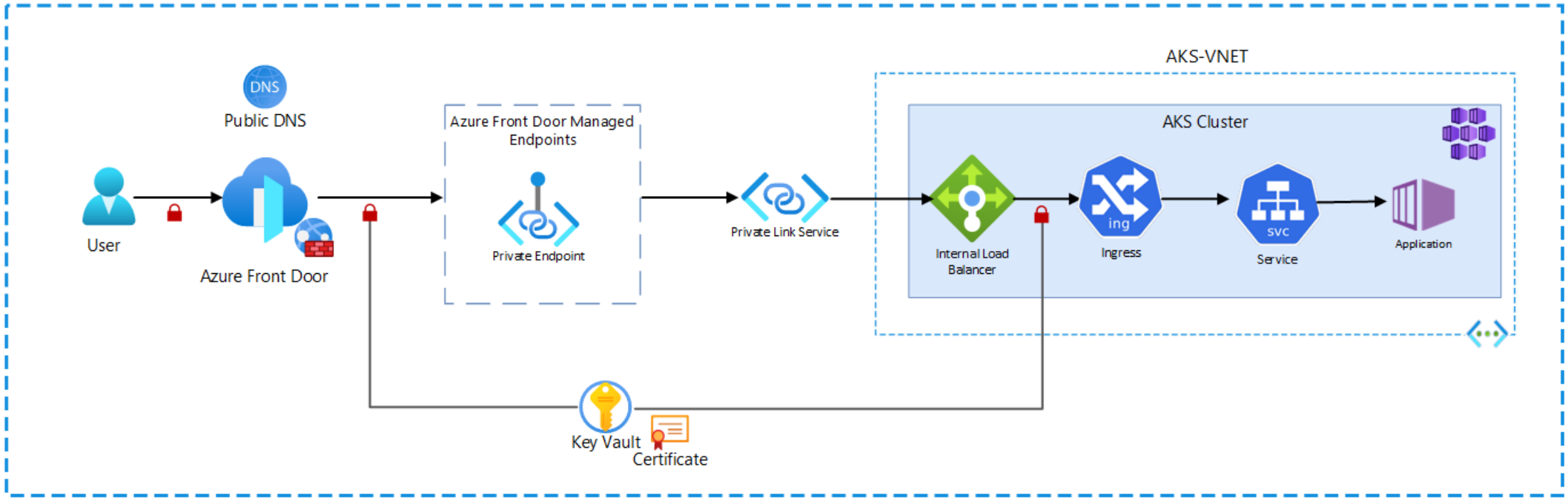
Front Door and Private Link Service (PLS)

- Front Door Premium can connect to a backend application via Azure Private Link Service (PLS).
- The backend system can be hosted in a VNET or hosted as a PaaS service such as Web App or Storage Account.
- PLS removes the need for your origin to be accessed publicly.
- When you enable Private Link to your origin in Azure Front Door Premium, Front Door creates a private endpoint on your behalf from a Front Door managed regional private network.
- You need to explicitly approve Front Door private endpoint request at the origin.
- If you deploy a private origin using Front Door Premium and the PLS, TLS/SSL offload is fully supported.

Private origins in Azure via private link service



Exposing AKS apps using Front Door and PLS

AKS creates an internal Load Balancer and Private Link Service through Kubernetes service annotations inside cluster VNET.

Then, Front Door can create a Private Endpoint in its own managed VNET to connect to the PLS of the cluster.